A Major Project Report
(Phase-I)

On

Intelligent Traffic Management System using Computer Vision

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**Intelligent Traffic Management System using Computer Vision**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Dr. Lalit Mohan Gupta**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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Table of Contents

SL. No	Section	Page no.
	Abstract	1
1	Problem Definition	2-3
2	Literature Survey	3-7
3	Objectives	7
4	Scope	8
5	Feasibility Study	9-11
	References	11-13
	Plagiarism Report	

ABSTRACT

Traffic congestion is becoming a serious problem in fast-growing cities these days. The growth in traffic volume, limited road capacity and fixed timing of traffic signals can cause long queues, increased waiting time and inefficient utilisation of road intersections. Traditional traffic lights generally use pre-set timing schedules and may not properly react to fast changes in traffic conditions. The proposal of this project, Intelligent Traffic Management System Using Computer Vision, is to build a software-based system that can evaluate traffic-camera footage and make intelligent traffic-management recommendations. The system uses computer vision for vehicle detection and classification, tracking of vehicle movement, vehicle count calculation and traffic density estimation at an intersection. Based on the traffic, system can propose suitable signal timings for different directions of traffic. The vehicle detection component can be implemented using the YOLO family of object detection models. The original YOLO research proved that object detection can be treated as a single process and achieved real-time processing at around 45 frames per second for its base model . A multi-object tracking method such as DeepSORT can be used to track vehicle identities across video frames. DeepSORT incorporated appearance information into the tracking process and demonstrated a 45% decrease in identity switches compared to the original SORT in their experiments . The traffic-control component is based on the research on adaptive traffic-signal control. IntelliLight has demonstrated the use of deep reinforcement learning for traffic-light control and tested the approach with large-scale real traffic data collected from surveillance cameras . At the beginning, the proposed system will operate as a software prototype on recorded or publicly available traffic videos. It won't just control physical traffic signals, but will provide traffic analysis and signal recommendations via a dashboard. The project will thus show how computer vision and intelligent decision-making can be combined to support adaptive traffic management.

PROBLEM DEFINITION

1. Background

Traffic intersections are one of the most important elements in the urban transportation system. When the traffic load increases, the traditional approach of using fixed-time traffic lights at the intersections fails to deliver the much-needed priority to the roads with a higher density of waiting cars.

For instance, if there are 30 cars waiting at one road and only 5 cars at the other road, the fixed-time traffic lights will give the same amount of priority to the roads with different traffic densities.

The challenge is especially prominent during:

- Rush hours
- School hours
- Office hours
- After accidents
- During maintenance
- Heavy rains
- Emergency cases
- Uneven traffic density
- Special occasions

A computer vision system can be used to track the traffic densities constantly and turn the visual data into information that can be used to update the traffic light policy in real-time.

2. Problem Statement

The project aims to design an integrated computer-vision intelligent traffic management system that can

I. Detect vehicles in traffic-camera videos
II. Classify vehicles
III. Track vehicles
IV. Count vehicles in different traffic directions
V. Estimate the traffic density
VI. Recognize congestion
VII. Identify emergency vehicles if present
VIII. Recommend adaptive signal timings
IX. Display traffic information

The final objective is to reduce the reliance on conventional time-based signal systems and recommend signals based on actual traffic observations.

3. Existing System

Most conventional traffic management systems are based on

- Conventional signal timing
- Manual control
- Sensors
- Human operators
- Customized rules

These systems can be effective for some traffic scenarios. However, research on IntelliLight shows that conventional signal control based on fixed-time plans has limitations in adapting to traffic conditions .

4. Proposed System

The proposed system can process traffic videos in the following pipeline:

Traffic Video → Vehicle Detection → Tracking → Counting → Density Estimation → Congestion Recognition → Signal Recommendation → Dashboard

which can allow the system to provide traffic information rather than relying on human observation.

LITERATURE SURVEY

1. Introduction

Computer Vision and Machine Learning have seen increasing application in intelligent transportation systems. The related research can be generally categorised into vehicle detection, tracking and traffic state estimation, and adaptive traffic control.

In the context of this project, the relevant research includes YOLO for object detection, DeepSORT for tracking, and reinforcement learning-based approaches such as IntelliLight and PressLight for signal control.

2. YOLO for Vehicle Detection

Redmon et al. proposed You Only Look Once (YOLO) in 2016 [1]. The research introduced a unified model that can predict object bounding boxes and class probabilities.

The main advantage of You Only Look Once is its speed: the original paper reported that the base model achieved approximately 45 FPS, while Fast YOLO achieved about 155 FPS in the experiment [1].

This is critically important for traffic-management applications, since video typically consists of dozens of frames per second that need to be processed.

For the proposed project, YOLO can be used for detecting the following objects:

- Cars
- Buses
- Trucks
- Motorcycles
- Other vehicles

The detection module can provide bounding boxes for the found objects in a video, which can then be passed to the tracking module.

The potential limitation of YOLO is that it may have localization errors, especially for small objects. In the context of traffic cameras, distant vehicles may appear as small objects in the field of view of the camera, which may reduce detection accuracy. Later versions of YOLO address this limitation by achieving better balance between speed and accuracy.

1. DeepSORT for Vehicle Tracking

Vehicle detection alone is not sufficient for traffic analysis, since it does not provide any information on whether the same vehicle is detected in two consecutive frames.

Wojke et al. proposed DeepSORT in 2017 [2]. The research introduced an approach to visual object tracking that extended the SORT framework by adding appearance information.

The paper reported that using appearance information reduced the number of identity switches by approximately 45% in the experiment [2].

This is critically important for the proposed traffic-management system, since visual object tracking is necessary to :

- Count vehicles
- Determine the traffic direction
- Estimate the density
- Carry out other analyses

For example, if a car appears in 20 consecutive frames, it should be counted as one vehicle, not 20. The potential limitation of DeepSORT is that it adds significant complexity over simpler tracking algorithms.

4. IntelliLight for Adaptive Traffic Control

Wei et al. proposed IntelliLight: A Reinforcement Learning Approach for Intelligent Traffic Light Control in 2018 [3].

The paper is focused on the limitations of conventional traffic-light systems and proposed

deep reinforcement learning for adaptive signal control. The researchers evaluated their approach on a large-scale real-world traffic dataset captured by surveillance cameras [3].

The general idea of IntelliLight is to represent the traffic state as the system's state, and use a reinforcement learning agent to choose the traffic signal as the agent's action, with rewards defined based on traffic performance.

The paper is directly relevant to the proposed project, since the traffic analysis module can estimate the traffic state for IntelliLight's agent. However, reinforcement learning requires significant experience for training, and for the initial version of the proposed project, a rule-based signal recommendation algorithm may be used with potential extension to reinforcement learning in later stages.

5. PressLight

PressLight is another approach to reinforcement learning-based traffic signal control. It uses the Max Pressure concept for traffic control, and applies reinforcement learning to estimate the max pressure.

The general idea is that traffic movements with the highest pressure should be prioritized when assigning traffic signal phases.

This concept may be applicable to the proposed system, since the traffic density estimated by computer vision can serve as a proxy for traffic demand. For example, the following rough estimation can be used:

Direction #	Vehicles	Pressure
North	25	High
South	21	High
East	7	Low
West	5	Low

According to this estimation, the traffic signal should prioritize north-south traffic over east-west traffic. This is important since it provides a link between machine learning and conventional traffic engineering.

6. CoLight

CoLight extends the concept of adaptive traffic signal control to multiple intersections. The key idea is that adjacent intersections impact each other, and therefore, a coordinated approach to traffic control is necessary.

It is relevant to the proposed project, since the initial version can be focused on a single intersection, but later versions can consider multiple intersections. The main limitation is that it adds significant complexity to the system.

7. CityFlow

CityFlow is a traffic simulation environment for large-scale city traffic. The main advantage of simulation is that it allows testing and comparing different traffic control policies.

For the proposed project, a traffic simulator can be used to test signal recommendation algorithms before proceeding to real-world traffic videos.

8. Literature Comparison

Paper Year Area Contribution Relevance

YOLO [1] 2016 Object Detection Real-time object detection Vehicle detection

DeepSORT [2] 2017 Object Tracking Improved accuracy Vehicle tracking

IntelliLight [3] 2018 Traffic Control RL-based adaptive signal control Signal recommendation

PressLight [4] 2019 Traffic Control RL-based max pressure estimation Traffic-demand estimation

CoLight [5] 2019 Network-level control multi-intersection cooperation Future work

CityFlow [6] 2019 Simulation Environment for testing RL algorithms

9. Research Findings

Based on the review of existing research, the following findings can be concluded:

1. Modern computer vision methods allow real-time object detection that can be used for traffic management. For example, YOLO showed that a single neural network can process video in real time [1].

2. Visual object tracking is a necessary step for processing video in traffic management applications. It can be achieved with DeepSORT, which improves accuracy by using appearance information [2].

3. Traffic control can be improved by adapting signal timings based on the current traffic state rather than following a fixed schedule. This can be achieved with IntelliLight, which applies reinforcement learning to traffic signal control [3].

4. Research such as PressLight and CoLight shows that machine learning can be integrated with conventional traffic engineering principles.

10. Limitations of Existing Research

Based on the review of literature, the following limitations can be identified:

1. Some detection algorithms are computationally expensive.

2. Tracking accuracy can be impacted by occlusion.

3. Detection accuracy can be affected by lighting conditions.
4. Traffic datasets may not be available for all locations.
5. Reinforcement learning may be difficult to apply in practice.
6. Many traffic research papers are based on simulation.
7. Simulation results may not generalize well to practice.
8. Multi-intersection approaches are more complex than single-intersection approaches.
9. Indian mixed traffic presents unique challenges.

11. Research Gap

The review of literature shows that computer vision, visual tracking, and reinforcement learning have been applied to traffic management. However, there is a research gap in developing an integrated solution that combines these areas. Specifically, the proposed project addresses the following research gap:

To develop an integrated software system that can utilize computer vision to estimate traffic information and apply reinforcement learning to recommend adaptive traffic signal timings.

As a starting point, the proposed system will focus on a single intersection and recorded traffic videos, which will allow addressing a practical problem at an appropriate scale.

OBJECTIVES

1. Main Objective

The objective of the project is to design an intelligent traffic management software system that can utilise computer vision to analyse traffic and recommend adaptive signal timings.

2. Specific Objectives

I. Detect vehicles in traffic-camera videos
II. Classify vehicles
III. Track vehicles
IV. Count vehicles in different directions
V. Estimate the traffic density
VI. Recognise congestion
VII. Recognise emergency vehicles
VIII. Recommend appropriate traffic signal timings
IX. Display traffic information
X. Evaluate the performance of detection, tracking and traffic recommendation modules

SCOPE

1. Included Scope

The scope of the project includes the following:

- Vehicle Detection:
  Use a deep-learning model to detect vehicles in traffic videos.
- Vehicle Classification:
  Classify detected vehicles into different categories.
- Vehicle Tracking:
  Track vehicles in video frames.
- Traffic Counting:
  Count detected vehicles in different directions.
- Density Estimation:
  Estimate the traffic density based on the number of vehicles and their arrangement.
- Congestion Recognition:
  Recognize congestion levels.
- Signal Recommendation:
  Recommend traffic signal timings based on congestion levels.
- Emergency Vehicle Detection:
  Recognize emergency vehicles if present.
- Dashboard:
  Provide the following information:
    - Number of vehicles
    - Traffic density
    - Congestion level
    - Vehicle categories
    - Recommended signal timings
    - Emergency vehicle information

2. Excluded Scope

The following aspects are outside of the scope of the project:

- Physical traffic lights
- Direct traffic signal control
- Roadside cameras
- City-wide traffic management
- Government traffic management systems
- Autonomous enforcement
- Large-scale traffic management

The project will focus on a computer vision software application.

FEASIBILITY STUDY

1. Technical Feasibility

The project is technically feasible, since the required technologies already exist. Specifically,

Python can be used since it provides extensive machine-learning and computer-vision libraries.

OpenCV can be used for

- Reading and processing videos
- Extracting video frames
- Drawing bounding boxes
- Defining regions of interest
- Plotting tracking results
- Saving the results as videos

YOLO can be used for real-time object detection, which is critically important for processing videos. The original YOLO research demonstrated its ability to run in real time [1].

DeepSORT can be used for visual object tracking, which is necessary for vehicle counting. The potential limitation is additional complexity, but it is justified by the need for accuracy. The paper reported that DeepSORT reduced the number of identity switches by approximately 45% [2].

A database can be used for storing traffic data, such as

- The number of vehicles
- Density
- Time
- Congestion level
- Recommended signal timings

A dashboard can be built with Streamlit, Flask or another web-application framework.

2. Economic Feasibility

The project is economically feasible, since most required technologies are open source or freely available. Specifically,

Resource Availability

Python Free

OpenCV Open source

YOLO Open source

NumPy Free

Pandas Free

Streamlit Free

Public datasets Available for research

Traffic simulator SUMO / CityFlow

Hardware Existing computer

No physical traffic-light infrastructure is needed for the prototype.

3. Operational Feasibility

The system will be easy to operate since it will have a dashboard that will allow the operator to

Current Traffic → Vehicle Count → Density → Congestion → Recommended Signal

For example, the following information can be presented to the operator:

Direction # Density Recommendation

North 28 High Increase green light duration

South 24 High Increase green light duration

East 8 Low -

West 5 Low -

This way, the operator can make decisions based on the information provided by the system, rather than estimating the traffic state manually.

4. Schedule Feasibility

The project can be completed within the timeframe since it can be prioritized in stages:

Stage 1: Problem definition and literature survey

Stage 2: Dataset collection and preparation

Stage 3: Vehicle detection

Stage 4: Vehicle tracking

Stage 5: Vehicle counting and density estimation

Stage 6: Congestion recognition

Stage 7: Signal recommendation

Stage 8: Dashboard development

Stage 9: Testing and performance evaluation

Stage 10: Documentation and presentation

5. Risk Analysis

Risk	Impact	Mitigation
Poor lighting	Poor detection	Use different training data
Heavy traffic	Poor tracking	Use a better tracker
Small vehicles	Missed detection	Increase the resolution
Low FPS	Increased delay	Use a faster model

GPU limitations	Slow training	Use a lighter model/train on cloud
Weather	Poor performance	Test under different conditions
Incorrect counting	Incorrect density estimation	Validate tracking and line crossing
RL complexity	Hard to implement	Start with a rule-based controller

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Rajat Mishra

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ABSTRACT

Traffic congestion is becoming a serious problem in fast-growing cities these days. The growth in traffic volume, limited road capacity, and fixed timing of traffic signals can cause long queues, increased waiting time and inefficient utilization of road intersections. Traditional traffic lights generally use pre-set timing schedules and may not properly react to fast changes in traffic conditions. The proposal of this project, Intelligent Traffic Management System Using Computer Vision, is to build a software-based system that can evaluate traffic-camera footage and make intelligent traffic-management recommendations. The system uses computer vision for vehicle detection and classification, tracking of vehicle movement, vehicle count calculation, and traffic density estimation at an intersection. Based on the traffic, the system can propose suitable signal timings for different directions of traffic. The vehicle detection component can be implemented using the YOLO family of object detection models. The original YOLO research proved that object detection can be treated as a single process and achieved real-time processing at around 45 frames per second for its base model. A multi-object tracking method such as DeepSORT can be used to track vehicle identities across video frames. DeepSORT incorporated appearance information into the tracking process and demonstrated a 45% decrease in identity switches compared to the original SORT in their experiments. The traffic-control component is based on the research on adaptive traffic-signal control. IntelliLight has demonstrated the use of deep reinforcement learning for traffic-light control and tested the approach with large-scale real traffic data collected from surveillance cameras. At the beginning, the proposed system will operate as a software prototype on recorded or publicly available traffic videos. It won't just control physical traffic signals but will provide traffic analysis and signal recommendations via a dashboard. The project will thus show how computer vision and intelligent decision-making can be combined to support adaptive traffic management.